

Additivity of spherical volume under free products in dimensions at least four

Abstract

We prove that integral-current spherical volume is additive on integral homology classes induced from the factors of a free product of countable groups in every dimension at least four. Consequently, it is additive under connected sum of closed connected oriented manifolds in these dimensions, answering the corresponding part of Song’s additivity question (Sém. Théor. Spectr. Géom., 2021–2022). The proof constructs integral transfers indexed by the vertices of the Bass–Serre tree. At each vertex, a shadow map replaces the coordinates in each branch by their norm. The sum of the n -Jacobians of these maps is at most one for $n \geq 4$. This joint estimate recovers the two factor classes with combined mass no greater than that of the original cycle; separate use of the group retractions does not provide such a bound. To apply the transfers to every competitor in the defining infimum, we establish mass-sharp approximation of cycles by liftable finite-support polyhedral cycles and a relative filling comparison. Both statements hold in the presence of finite stabilizers and preserve integer multiplicities.

Note. This paper was generated entirely by AI, including MiMo, using an automated research pipeline developed by Chenghua Liu and Hanyu Li.

1 Introduction

Spherical volume was introduced by Besson, Courtois, and Gallot in their work on volume entropy and locally symmetric spaces [2, 3]. Song’s integral-current formulation [10] assigns a spherical volume to an integral homology class of any countable group: it minimizes the mass of current representatives in a quotient of a Hilbert sphere. For a connected sum, the fundamental group is a free product and the induced fundamental class is the sum of the two factor classes. These splittings give an upper bound for spherical volume. Additivity asks whether a cycle in the larger quotient can have less mass than the sum of the two separate infima.

Song poses this question for closed oriented manifolds of dimension at least three [10, Question 10]. His computation in dimension three [10, Theorem 5.1] already expresses spherical volume in terms of the hyperbolic pieces of the prime and JSJ decompositions, and hence implies connected-sum additivity in that dimension. We prove the formula in every dimension at least four, more generally for arbitrary integral homology classes in the factors of a free product. The separate question in [10, Question 10] concerning decomposition of spherical Plateau solutions is not addressed here.

There are two useful comparisons with earlier work. For the classical map-based spherical volume, Brunnbauer proves homological invariance and invariance under connected sum with a simply connected manifold [4, Theorem 1.1 and Corollary 8.3]. The latter does not give a sum formula when both summands have nonzero spherical volume. Moreover, the classical and integral-current definitions should not be identified without hypotheses; Song discusses their comparison in [10, Section 3.5]. For simplicial volume, Gromov’s connected-sum formula holds in dimensions at least three [7, Section 3.5]. Here the quantity to be controlled is geometric mass with integer multiplicities, rather than the real simplicial seminorm. Our proof uses a joint Jacobian inequality and integral filling comparisons.

For a countable group G , set

$$S_G = S(\ell^2(G; \mathbb{R})), \quad X_G = S_G/G,$$

where $S(H)$ denotes the unit sphere of a real Hilbert space H , and G acts by its left regular representation, $(\gamma u)(g) = u(\gamma^{-1}g)$. We use the angular metric

$$d_{S_G}(u, v) = \arccos \langle u, v \rangle, \quad d_{X_G}([u], [v]) = \inf_{\gamma \in G} d_{S_G}(u, \gamma v).$$

The angular metric is the round geodesic metric whenever the sphere has positive dimension. This convention also covers the trivial group.

We write $\mathbf{I}_k^c(X)$ for the integral k -currents with compact support in a metric space X , and $H_k^{\text{IC},c}(X)$ for their homology. The mass and mass measure of a current C are denoted by $\mathbf{M}(C)$ and $\|C\|$. We use the pushforward, boundary, homotopy, and area formulas for metric currents from [1]. All fillings below have compact support and integer multiplicities.

For $k \geq 1$, Song defines a natural map

$$\hat{\pi}_G : H_k(G; \mathbb{Z}) \longrightarrow H_k^{\text{IC},c}(X_G).$$

For infinite G , it is obtained from the inclusion of the quotient of the free locus of S_G , a classifying space for G ; for finite G , it is zero by definition. Its spherical volume is

$$\text{SphereVol}_G(h) = \inf \{ \mathbf{M}(C) : C \in \mathbf{I}_k^c(X_G), \partial C = 0, [C] = \hat{\pi}_G(h) \}. \quad (1)$$

Throughout, we use the definitions and normalization of [10, Sections 3.1–3.3].

Theorem 1.1. *Let A, B be countable groups, let $G = A * B$, and let $i_A : A \rightarrow G$ and $i_B : B \rightarrow G$ be the inclusions. For every $n \geq 4$, $\alpha \in H_n(A; \mathbb{Z})$, and $\beta \in H_n(B; \mathbb{Z})$,*

$$\text{SphereVol}_G((i_A)_*\alpha + (i_B)_*\beta) = \text{SphereVol}_A(\alpha) + \text{SphereVol}_B(\beta).$$

For a closed connected oriented manifold M , let $h_M \in H_n(\pi_1(M); \mathbb{Z})$ be the image of its fundamental class under a classifying map, and write $\text{SphereVol}(M) = \text{SphereVol}_{\pi_1(M)}(h_M)$.

Corollary 1.2. *If M, N are closed connected oriented manifolds of the same dimension $n \geq 4$, then*

$$\text{SphereVol}(M \# N) = \text{SphereVol}(M) + \text{SphereVol}(N).$$

The upper bound follows from two properties of (1). Addition of currents gives subadditivity. For a homomorphism $\theta : F \rightarrow G$, the norm-folding map

$$\Theta_\theta(u)(g) = \left(\sum_{x \in \theta^{-1}(g)} u(x)^2 \right)^{1/2} \quad (2)$$

is θ -equivariant and distance nonincreasing; an empty fiber has value zero. Its quotient map takes $\hat{\pi}_F(h)$ to $\hat{\pi}_G(\theta_*h)$ [10, Section 3.4.1]. Hence

$$\text{SphereVol}_G(\theta_*h) \leq \text{SphereVol}_F(h). \quad (3)$$

Applying this to the two inclusions proves the required upper bound.

The lower bound requires more than the two retractions $G \rightarrow A, B$. For example, choose distinct $a_0, \dots, a_4 \in A$ and distinct $b_0, \dots, b_4 \in B$. On the positive spherical four-simplex with coordinate vertices $\delta_{a_j b_j}$, both retraction foldings are coordinate isometries. Their four-Jacobians add to two, not one. We instead sum shadow maps indexed by the vertices of the Bass–Serre tree. They induce transfers to the two factor quotients and satisfy a joint mass bound with constant one.

The dimension restriction enters when the tree is cut across an edge. The two complementary foldings have a common radial space of dimension at most two. In a tangent n -plane with $n \geq 4$, a two-plane remains orthogonal to that space; on it, the differentials are complementary projections. A determinant inequality bounds the sum of the full Jacobians by one. Induction over a finite tree then gives the transfer estimate.

There is also an approximation issue: the defining infimum includes all compactly supported integral cycles, including cycles meeting points with finite stabilizers. We prove that they can be replaced by liftable finite-support polyhedral cycles with arbitrarily small mass increase, and that polyhedral cycles which bound integral currents have polyhedral fillings. Song’s approximation lemma [10, Lemma 1.6] treats locally spherical manifolds; the argument here supplies the finite-isotropy and relative filling statements needed for the transfer on the full quotient.

2 A Jacobian inequality for complementary foldings

For a linear map D on an n -dimensional Euclidean space E , set

$$\text{Jac}_n(D|E) = \sqrt{\det((D|E)^*(D|E))}.$$

For a Lipschitz map on a rectifiable set, the same notation refers to its approximate tangential differential. All sphere metrics in this section are round metrics.

Lemma 2.1 (Two norm foldings). *Let H_1, H_2 be finite-dimensional real Hilbert spaces. Define*

$$\begin{aligned} L : S(H_1 \oplus \mathbb{R} \oplus H_2) &\longrightarrow S(H_1 \oplus \mathbb{R}), & L(x, z, y) &= (x, \sqrt{z^2 + \|y\|^2}), \\ R : S(H_1 \oplus \mathbb{R} \oplus H_2) &\longrightarrow S(H_2 \oplus \mathbb{R}), & R(x, z, y) &= (y, \sqrt{z^2 + \|x\|^2}). \end{aligned}$$

If $n \geq 4$, then at almost every tangent n -plane E of a rectifiable n -current in the domain,

$$\text{Jac}_n(L|E) + \text{Jac}_n(R|E) \leq 1. \tag{4}$$

Proof. Cauchy–Schwarz shows that folding a block to its norm increases the inner product of any two unit vectors. Since arccos is decreasing, both maps are contractions.

Suppose first that $x \neq 0$ and $y \neq 0$. At $u = (x, z, y)$, put

$$U_x = \{(\xi, 0, 0) : \langle \xi, x \rangle = 0\}, \quad U_y = \{(0, 0, v) : \langle v, y \rangle = 0\},$$

and

$$W = u^\perp \cap \text{span}\{(x, 0, 0), (0, 1, 0), (0, 0, y)\}.$$

Then $T_u S = U_x \oplus U_y \oplus W$ orthogonally and $\dim W \leq 2$. On $U_x \oplus U_y$, the differentials of L and R are the two complementary orthogonal projections, with their images identified isometrically with U_x and U_y .

For an n -plane $E \subset T_u S$, choose a two-plane $E_0 \subset E \cap W^\perp$. In an orthonormal basis e_1, e_2 of E_0 , let D_L, D_R be the Gram matrices of dL and dR . Thus

$$D_L, D_R \geq 0, \quad D_L + D_R = I_2.$$

Complete e_1, e_2 to an orthonormal basis of E . The norm of an exterior product is at most the product of the norms of its factors, and $\|dL\|, \|dR\| \leq 1$. Consequently,

$$\text{Jac}_n(L|E) \leq \sqrt{\det D_L}, \quad \text{Jac}_n(R|E) \leq \sqrt{\det D_R}.$$

If $\lambda_1, \lambda_2 \in [0, 1]$ are the eigenvalues of D_L , then

$$\sqrt{\det D_L} + \sqrt{\det D_R} = \sqrt{\lambda_1 \lambda_2} + \sqrt{(1 - \lambda_1)(1 - \lambda_2)} \leq 1$$

by Cauchy–Schwarz. This proves (4) at these points.

For the almost-everywhere assertion, partition the sphere according to which coordinate blocks vanish. At almost every point of the portion of a rectifiable set in a zero level set of a Lipschitz coordinate function, the tangential derivative of that function is zero. Therefore the approximate tangent is contained in the tangent space of the corresponding stratum. If $x = 0$ on that stratum, L is constant there and R is a contraction; the case $y = 0$ is symmetric. The case $x = y = 0$ has no positive-dimensional tangent. These cases complete the proof. In particular, the estimate holds pointwise for every tangent plane to each smooth coordinate stratum. $\square$

Definition 2.2 (Vertex shadows). Let T be a finite tree with at least one edge. Its edges index the coordinates of $\ell^2(E(T); \mathbb{R})$. For a vertex v , deleting v partitions the edge midpoints into branches, one for each incident edge $e \in E_v(T)$. Define

$$p_v : S(\ell^2(E(T); \mathbb{R})) \longrightarrow S(\ell^2(E_v(T); \mathbb{R})), \quad p_v(u)(e) = \left(\sum_{a \text{ in branch } e} u(a)^2 \right)^{1/2}. \quad (5)$$

Lemma 2.3 (Tree Jacobian bound). *For $n \geq 4$, at almost every tangent n -plane E of a rectifiable current in $S(\ell^2(E(T); \mathbb{R}))$,*

$$\sum_{v \in V(T)} \text{Jac}_n(p_v|E) \leq 1. \quad (6)$$

Proof. If T is a star, every leaf shadow is constant and the center shadow is coordinatewise absolute value, a contraction. This also covers a tree with a single edge.

Otherwise choose an edge e with at least one other edge on each side. Cut at e and retain it as a terminal edge in both resulting trees T_L, T_R . Write a vector as (x, z, y) , where z is the coordinate of e , and x, y are the coordinates strictly on the two sides. Folding the opposite side together with z into the terminal coordinate gives the maps L, R of Lemma 2.1.

Each original left-side shadow is a shadow of T_L composed with L ; the analogous statement holds on the right. The additional terminal vertices have constant shadows. If $dL|E$ has rank n , the Jacobian chain rule and induction on T_L give

$$\sum_{v \text{ on the left}} \text{Jac}_n(p_v|E) \leq \text{Jac}_n(L|E).$$

If its rank is smaller, every term on the left is zero. The same argument applies to R . The pointwise coordinate-stratum version of Lemma 2.1 permits induction on the image tangent planes, including when some folded coordinates vanish. Hence

$$\sum_v \text{Jac}_n(p_v|E) \leq \text{Jac}_n(L|E) + \text{Jac}_n(R|E) \leq 1.$$

$\square$

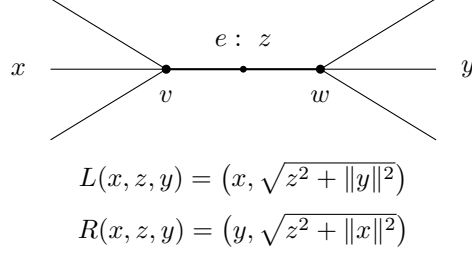


Figure 1: Cutting an edge produces two complementary foldings. The coordinate of the cut edge is retained in both smaller trees.

3 Integral transfers on the Bass–Serre tree

Let $G = A * B$. Its Bass–Serre tree [9, Chapter I] has vertex set $G/A \sqcup G/B$ and an edge labelled g joining gA to gB for each $g \in G$. Place the coordinate indexed by g at that edge’s midpoint.

For an A -vertex $v = g_v A$, label its incident edges by $a \in A$ via $g_v a$, and let $\mathcal{B}(v, a) \subset G$ be the edge midpoints in the corresponding branch. Define

$$\psi_v(u)(a) = \left(\sum_{g \in \mathcal{B}(v, a)} u(g)^2 \right)^{1/2}, \quad q_v : S_G \longrightarrow X_A, \quad q_v(u) = [\psi_v(u)]. \quad (7)$$

Changing g_v changes ψ_v by the left regular action of A , so q_v does not depend on the representative. Define $q_v : S_G \rightarrow X_B$ in the same way at a B -vertex. These maps are contractions and satisfy

$$q_{\gamma v}(\gamma u) = q_v(u) \quad (\gamma \in G). \quad (8)$$

A *liftable finite-support polyhedral current* in X_G is a finite integer sum of projected spherical geodesic simplices, each lying in an open hemisphere of a finite coordinate sphere. We require the quotient map to be injective on each lifted simplex and locally isometric along its relative interior. Subdivision and cancellation are understood at the level of currents. Denote their group in degree k by $\mathcal{P}_k(X_G)$. The integration current on an oriented lifted simplex $\tilde{\sigma}$ is written $[\![\tilde{\sigma}]\!]$.

The following approximation statement will be proved in Section 5. It permits the transfer to be defined on finite chains without restricting the competitors or homology comparisons in (1).

Lemma 3.1 (Approximation and fillings). *Let G be countable and $k \geq 1$.*

- (i) *For every $C \in \mathbf{I}_k^c(X_G)$ with $\partial C = 0$ and every $\varepsilon > 0$, there exist $P \in \mathcal{P}_k(X_G)$ and $D \in \mathbf{I}_{k+1}^c(X_G)$ such that*

$$\partial P = 0, \quad C - P = \partial D, \quad \mathbf{M}(P) \leq \mathbf{M}(C) + \varepsilon.$$

- (ii) *If $Z \in \mathcal{P}_k(X_G)$ bounds a current in $\mathbf{I}_{k+1}^c(X_G)$, then $Z = \partial Q$ for some $Q \in \mathcal{P}_{k+1}(X_G)$, after common subdivision. No bound on $\mathbf{M}(Q)$ is asserted.*

For a projected oriented simplex σ with lift $\tilde{\sigma}$, define in positive degrees

$$\mathcal{T}_A(\sigma) = \sum_{v \in G/A} (q_v)_\# [\![\tilde{\sigma}]\!], \quad \mathcal{T}_B(\sigma) = \sum_{v \in G/B} (q_v)_\# [\![\tilde{\sigma}]\!], \quad (9)$$

and extend by integer linearity. The images need not be polyhedral.

Proposition 3.2. *The expressions (9) define homomorphisms*

$$\mathcal{T}_A : \mathcal{P}_k(X_G) \longrightarrow \mathbf{I}_k^c(X_A), \quad \mathcal{T}_B : \mathcal{P}_k(X_G) \longrightarrow \mathbf{I}_k^c(X_B) \quad (k \geq 1),$$

independently of lifts and polyhedral presentations. For $k \geq 2$,

$$\partial \mathcal{T}_A(P) = \mathcal{T}_A(\partial P), \quad \partial \mathcal{T}_B(P) = \mathcal{T}_B(\partial P). \quad (10)$$

For $k \geq 4$,

$$\mathbf{M}(\mathcal{T}_A P) + \mathbf{M}(\mathcal{T}_B P) \leq \mathbf{M}(P). \quad (11)$$

Proof. A finite coordinate set containing the vertices of $\tilde{\sigma}$ contains every point of that simplex. Take the finite subtree spanned by the corresponding edges. At a vertex outside this subtree, all those coordinates lie in one branch, so the shadow is a constant coordinate unit vector. Its pushforward in positive degree vanishes. Both sums in (9) are therefore finite, and produce integral currents with compact support.

Changing a lift by $\gamma \in G$ permutes the summands by (8). Subdivision is compatible with Lipschitz pushforward. To compare different presentations, take common refinements of their lifted spherical polyhedra and their relevant translates. Only finitely many translates occur: two coordinate spheres supported on F, F' can meet after translation only if $F \cap \gamma F' \neq \emptyset$, which restricts γ to the finite set $F(F')^{-1}$. The intersections are spherical polyhedra and admit common subdivisions. On each resulting piece the quotient lifts agree up to translation. Equal currents consequently have equal transfers after cancellation of their integer multiplicities.

This argument also applies in an isotropy stratum. An element fixing a lifted piece pointwise permutes its vertex-indexed shadow summands. Coincident image currents retain their integer multiplicities: we sum over all vertices, not over stabilizer orbits with an averaging factor. Thus no division by a stabilizer order is involved.

For $k \geq 2$, the pushforward boundary formula applies simplex by simplex. Every face has positive dimension, so all the sums on faces are finite. The lift of a shared face may be changed by a group element, which only permutes the summands. This proves (10). No degree-zero transfer is needed.

For the mass estimate, use a presentation of P with disjoint relative interiors and integer multiplicities. On a lifted simplex, enlarge the coordinate set to the edges of its finite spanning subtree. The maps ψ_v are its vertex shadows followed by coordinate inclusions; q_v also includes a contracting quotient map. Lemma 2.3 and the area formula give

$$\begin{aligned} & \sum_{v \in G/A \sqcup G/B} \mathbf{M}((q_v)_\# [\tilde{\sigma}]) \\ & \leq \int_{\tilde{\sigma}} \sum_v \text{Jac}_k(q_v) d\mathcal{H}^k \leq \text{Vol}_k(\tilde{\sigma}). \end{aligned}$$

The lifted and quotient simplex volumes agree by the local isometry condition. Summing with absolute multiplicities, and allowing cancellation in the images, proves (11). $\square$

4 Additivity and connected sums

We first identify the transfers on cycles in the factor spheres. The boundary formula then identifies the transferred classes of every competitor, not just cycles with a particular geometric position.

Proof of Theorem 1.1. By Lemma 3.1(i), choose cycles $P_A \in \mathcal{P}_n(X_A)$ and $P_B \in \mathcal{P}_n(X_B)$ representing $\hat{\pi}_A(\alpha)$ and $\hat{\pi}_B(\beta)$. For a finite factor, choose the zero cycle. Let $j_A : X_A \rightarrow X_G$ and $j_B : X_B \rightarrow X_G$ be the maps induced by extension of coordinates by zero. They are injective contractions and are locally isometric along lifted simplices. Indeed, if two vectors supported on A are related by an element of G , that element belongs to A ; translates by elements outside A have disjoint supports and are at angular distance $\pi/2$. The same holds for B .

Set

$$P_0 = (j_A)_\# P_A + (j_B)_\# P_B.$$

Compatibility with the defining group-homology maps gives

$$[P_0] = \hat{\pi}_G((i_A)_* \alpha + (i_B)_* \beta).$$

Here extension by zero has the same effect on these classes as the folding map for the inclusion: the latter is extension by zero after coordinatewise absolute value, and absolute value preserves the image of $\hat{\pi}$ [10, Sections 3.4.1–3.4.2].

On S_A , the shadow at the vertex A is coordinatewise absolute value and every other shadow is constant. Writing $\mathcal{A}_A$ and $\mathcal{A}_B$ for the quotient absolute-value maps, we obtain

$$\mathcal{T}_A(P_0) = (\mathcal{A}_A)_\# P_A, \quad \mathcal{T}_B(P_0) = (\mathcal{A}_B)_\# P_B.$$

These cycles represent $\hat{\pi}_A(\alpha)$ and $\hat{\pi}_B(\beta)$, respectively.

Let $C \in \mathbf{I}_n^c(X_G)$ be any cycle in the class of P_0 and fix $\varepsilon > 0$. Lemma 3.1(i) gives a homologous $P \in \mathcal{P}_n(X_G)$ with $\mathbf{M}(P) \leq \mathbf{M}(C) + \varepsilon$. The cycle $P - P_0$ bounds a compact integral current, so part (ii) gives $Q \in \mathcal{P}_{n+1}(X_G)$ with $\partial Q = P - P_0$. Proposition 3.2 now implies

$$\begin{aligned} \mathcal{T}_A(P) - (\mathcal{A}_A)_\# P_A &= \partial \mathcal{T}_A(Q), \\ \mathcal{T}_B(P) - (\mathcal{A}_B)_\# P_B &= \partial \mathcal{T}_B(Q). \end{aligned}$$

In particular, $\mathcal{T}_A(P)$ and $\mathcal{T}_B(P)$ are admissible competitors for the two factor classes. Therefore

$$\begin{aligned} \text{SphereVol}_A(\alpha) + \text{SphereVol}_B(\beta) &\leq \mathbf{M}(\mathcal{T}_A P) + \mathbf{M}(\mathcal{T}_B P) \\ &\leq \mathbf{M}(P) \leq \mathbf{M}(C) + \varepsilon. \end{aligned}$$

Take the infimum over C and let $\varepsilon \downarrow 0$. The upper bound follows from subadditivity and (3). $\square$

Proof of Corollary 1.2. Put $A = \pi_1(M)$ and $B = \pi_1(N)$. Since $n \geq 3$, van Kampen's theorem gives $\pi_1(M \# N) = A * B$. The pinch map $M \# N \rightarrow M \vee N$ sends the oriented fundamental class to $[M] + [N]$. Composing with the classifying maps of the two summands gives

$$h_{M \# N} = (i_A)_* h_M + (i_B)_* h_N.$$

Apply Theorem 1.1. $\square$

Iteration gives the same formula for every finite connected sum in dimension at least four. In particular, adjoining a summand of zero spherical volume does not change the value. For closed oriented real-hyperbolic four-manifolds (M_j, g_j) with sectional curvature -1 , [10, Theorem 4.1] yields

$$\text{SphereVol}(M_1 \# \cdots \# M_r) = \frac{81}{256} \sum_{j=1}^r \text{Vol}(M_j, g_j).$$

Indeed, the metric normalization in that theorem is $(n-1)^2/(4n)$; for $n = 4$, the volume scaling factor is $(9/16)^2 = 81/256$.

5 Polyhedral approximation in spherical quotients

We prove Lemma 3.1 in two stages. First we move a compact set into a finite coordinate carrier by an equivariant deformation whose local Lipschitz constant is arbitrarily close to one. This deformation can fix a prescribed polyhedral boundary. We then prove mass-sharp approximation inside the carrier and compare its integral-current and simplicial homology. The group action is proper, but no local compactness of the full Hilbert-sphere quotient is assumed.

5.1 Finite carriers and relative deformation

For finite nonempty subsets $F_1, \dots, F_s \subset G$, set

$$\tilde{Y} = \bigcup_{\gamma \in G, 1 \leq j \leq s} S(\ell^2(\gamma F_j; \mathbb{R})), \quad Y = \tilde{Y}/G \subset X_G. \quad (12)$$

The spherical cross-polytope triangulations, with vertices $\pm \delta_g$, agree on intersections of coordinate spheres. They form a locally finite complex with finitely many G -orbits of simplices. For local finiteness, fix a point with a nonzero coordinate at g . Every sufficiently close point still has a nonzero coordinate there, and only finitely many sets γF_j contain g .

After spherical barycentric subdivision, an element preserving a simplex fixes it pointwise: its vertices are barycenters of a chain of faces of different dimensions, so each vertex must be fixed. Moreover, if two points of a closed barycentric simplex are equivalent under G , their minimal faces in that simplex are carried to one another. Face dimensions distinguish the vertices in the chain, so the two points are equal. Thus each quotient simplex has an injective lift. At an interior point, every stabilizer element fixes the lifted simplex pointwise; the quotient map is therefore locally isometric along it.

The quotient is a finite regular cell complex with spherical simplex cells; subdividing once more if needed gives a finite simplicial complex. Equip Y with the piecewise spherical length metric on each connected component. This metric and the inherited quotient metric are bi-Lipschitz equivalent on each component. Locally, a spherical chart and a gnomonic projection reduce the comparison to finitely many incident Euclidean polyhedral cones and a finite stabilizer. For a finite union of such cones, path distance within each local component is bounded by a constant times ambient distance, by the positive angles between the finitely many faces. The reverse bound follows from the definition of path length. Compactness gives the comparison on each component. There are finitely many components, and currents can be treated componentwise.

More precisely, these two metrics give the same mass to rectifiable currents in Y . Partition a rectifiable set by the relative open faces of the finite complex. At almost every point of each part, its tangent lies in that face. Along the face the two infinitesimal metrics agree, by the local isometry just established. The area formula proves equality of masses. We henceforth pass between these two realizations of a current supported in Y without changing its mass.

Any finite collection of liftable finite-support spherical polyhedra contained in Y can be included as a subcomplex after common refinement. Only finitely many translates of each relevant coordinate sphere meet any fixed coordinate sphere, since an intersection requires $F \cap \gamma F' \neq \emptyset$. Subdivide all these intersections equivariantly; there are finitely many quotient pieces.

Lemma 5.1. *Let $K \subset X_G$ be compact, and let $Z \subset K$ be the support of a liftable finite-support polyhedral current. For every $\eta > 0$, there is a map F on a neighborhood of K such that*

- (a) *F is locally $(1 + \eta)$ -Lipschitz;*
- (b) *$F(K)$ is contained in a carrier of the form (12);*
- (c) *$F|_K$ is homotopic to the inclusion by a Lipschitz homotopy on $[0, 1] \times K$ with compact image, fixing Z pointwise.*

The case $Z = \emptyset$ imposes no relative condition.

Proof. For infinite G , the matrix coefficients of the regular representation vanish at infinity:

$$\langle u, \gamma v \rangle \longrightarrow 0 \quad \text{as } \gamma \text{ leaves finite subsets of } G.$$

This follows for finitely supported vectors by disjointness of their supports, and for all vectors by approximation in ℓ^2 . The convergence is uniform when u, v range over fixed compact sets, by finite-net approximation. Consequently the action on S_G is proper and each stabilizer H_u is finite. A sufficiently small ball B_u about u has $\gamma B_u \cap B_u = \emptyset$ for $\gamma \notin H_u$, and its quotient is a chart B_u/H_u . For finite G , the same chart description follows by taking a ball smaller than the distances to the finitely many distinct translates of its center.

Choose finitely many such charts, with smaller concentric charts still covering K . In each larger chart, the lifts of the corresponding compact part of K are compact, because quotient maps by finite groups have compact inverse images of compact sets. Choose smooth cutoffs in the larger lifted balls, positive on the smaller balls, and translate them equivariantly. The cutoffs may be chosen as functions of the inner product with the center, hence invariant under its stabilizer. Normalizing their sum gives an equivariant partition of unity $\{\phi_j\}$ on an invariant neighborhood of the inverse image of K ; the index includes chart translates. Distinct translates within a chart family are disjoint, and there are only finitely many chart families. On a smaller invariant neighborhood,

$$\sum_j \|d\phi_j\| \leq C_0 \tag{13}$$

for a finite constant C_0 , since the normalizing denominator is bounded away from zero modulo G .

Fix $\delta > 0$. For each chart representative choose a finite-coordinate orthogonal projection P_j satisfying

$$\|P_j u - u\| \leq \delta$$

on the compact lifted part of K in its support. Enlarge the coordinate set to make it invariant under the finite stabilizer, and translate the projections equivariantly. In the relative case, also include the coordinate supports of every lifted polyhedral piece of Z meeting the chart. There are only finitely many such translates: the chart radius can be taken smaller than $\pi/2$, and uniform decay of matrix coefficients on each compact lifted polyhedron rules out all but finitely many translates meeting the chart. Thus

$$P_j u = u \quad \text{for } [u] \in Z \text{ whenever } \phi_j(u) \neq 0.$$

Define

$$R(u) = \sum_j \phi_j(u) P_j u, \quad \tilde{F}(u) = \frac{R(u)}{\|R(u)\|}.$$

After shrinking the invariant neighborhood, every active error is at most 2δ . Hence $\|R(u) - u\| \leq 2\delta$ and $\|R(u)\| \geq 1 - 2\delta$. Since $\sum_j d\phi_j = 0$,

$$DR(v) = \sum_j \phi_j P_j v + \sum_j d\phi_j(v)(P_j u - u).$$

The first operator has norm at most one, being a convex combination of orthogonal projections; (13) bounds the second by $2C_0\delta$. The derivative of normalization has norm $1/\|R(u)\|$, so the local Lipschitz constant of $\tilde{F}$ is at most

$$\frac{1 + 2C_0\delta}{1 - 2\delta}.$$

Choose $\delta < 1/2$ sufficiently small to make this at most $1 + \eta$.

The map and the normalized straight homotopy

$$\tilde{H}(t, u) = \frac{(1-t)u + tR(u)}{\|(1-t)u + tR(u)\|}$$

are equivariant and fix every lift of Z . Their denominators are bounded below by $1 - 2\delta$, so they descend to the stated map and homotopy. Their restrictions to the compact domains in the assertion are Lipschitz: local Lipschitz bounds give a uniform bound on close pairs, and compactness bounds the remaining pairs. The homotopy image is compact.

Only finitely many projections are active on each lifted neighborhood, so its image under $\tilde{F}$ lies in a single finite coordinate sphere. A finite cover of K now gives finitely many such coordinate sets modulo G , proving that $F(K)$ lies in a carrier of the form (12). $\square$

For $C \in \mathbf{I}_k^c(X_G)$ supported in K , the area formula gives

$$\mathbf{M}(F_{\#}C) \leq (1 + \eta)^k \mathbf{M}(C). \quad (14)$$

If $\partial C = 0$, the homotopy formula gives

$$F_{\#}C - C = \partial H_{\#}([0, 1] \times C).$$

If D has boundary supported in the prescribed set Z , then $\partial(F_{\#}D) = F_{\#}\partial D = \partial D$ exactly.

5.2 Mass-sharp approximation in a finite polyhedron

A neighborhood retraction onto a finite carrier need not have Lipschitz constant close to one. We instead use its derivative on the tangent planes of strict Euclidean approximants. Along a face containing the limiting tangent, every incident retraction piece restricts to the identity. This recovers the sharp mass constant despite possible singularities of the carrier.

Lemma 5.2. *Let Y be a finite spherical polyhedron with the piecewise spherical metric on each component, and let $k \geq 1$. Every integral k -cycle in Y is homologous in Y to a sequence of finite spherical polyhedral cycles whose masses converge to its mass. Moreover, a spherical polyhedral cycle which bounds an integral current in Y bounds a spherical polyhedral chain there, after subdivision.*

Proof. Work on one connected component at a time. Realize a triangulation of Y as a finite Euclidean polyhedron $K_0 \subset \mathbb{R}^d$, enlarging d if necessary so that $d > \dim Y$. The map $h : K_0 \rightarrow Y$ taking Euclidean barycentric coordinates to normalized spherical barycentric coordinates in each lifted simplex is a piecewise projective bi-Lipschitz homeomorphism. The maps agree on common faces and have smooth extensions on each closed simplex. For an integral cycle C in Y , put $T = (h^{-1})_{\#}C$.

We use the following explicit neighborhood retraction. Triangulate a Euclidean neighborhood of K_0 and subdivide so that K_0 is a full subcomplex. Let U be the open set where the sum of barycentric coordinates belonging to vertices of K_0 is positive. Deleting the other coordinates and renormalizing defines a retraction $r : U \rightarrow K_0$. On a compact neighborhood of K_0 contained in U , the map r is Lipschitz and piecewise projective. On each of finitely many pieces, $f = h \circ r$ has a smooth lift into a spherical simplex and maps affine simplices to spherical geodesic simplices, or to sets of smaller dimension, after subdivision.

The Euclidean approximation and deformation theorems [6], in the localized form of [5, Sections 4.1–4.2], give the following input. A compact integral cycle T admits integral polyhedral cycles T_i and compact integral currents S_i with

$$T - T_i = \partial S_i, \quad \mathbf{M}(S_i) \rightarrow 0, \quad \mathbf{M}(T_i) \rightarrow \mathbf{M}(T), \quad (15)$$

whose supports lie in arbitrarily small neighborhoods of $\text{spt } T$. The exact cycle condition can be obtained as follows. Localized polyhedral approximation first gives integral decompositions

$$T = P_i + R_i + \partial W_i, \quad \mathbf{M}(P_i) \leq \mathbf{M}(T) + o(1), \quad \mathbf{M}(R_i) + \mathbf{M}(W_i) = o(1).$$

Here P_i is polyhedral, and all supports approach $\text{spt } T$. Since $\partial R_i = -\partial P_i$ is polyhedral, the polyhedral-boundary version of the Euclidean deformation theorem, at mesh size a_i , gives

$$R_i = P'_i + U_i + \partial V_i,$$

where P'_i, U_i are integral polyhedral currents and, for a constant c_d depending only on the ambient dimension,

$$\begin{aligned} \mathbf{M}(P'_i) &\leq c_d \mathbf{M}(R_i), \\ \mathbf{M}(U_i) &\leq c_d a_i \mathbf{M}(\partial R_i), \\ \mathbf{M}(V_i) &\leq c_d a_i \mathbf{M}(R_i). \end{aligned}$$

Their supports lie within $c_d a_i$ of $\text{spt } R_i$. Choose $a_i \downarrow 0$ so that $a_i \mathbf{M}(\partial R_i) \rightarrow 0$, and set $T_i = P_i + P'_i + U_i$, $S_i = W_i + V_i$. Then $\partial T_i = 0$, $T - T_i = \partial S_i$, and the upper mass bound follows. Flat convergence and lower semicontinuity give the mass convergence in (15). The approximation and deformation operations use integer multiplicities throughout.

For all sufficiently large i , the supports of T_i, S_i lie in a fixed compact neighborhood contained in U . Define

$$C_i = f_{\#} T_i.$$

Each C_i is a finite integral spherical polyhedral cycle, and

$$C - C_i = \partial(f_{\#} S_i).$$

In particular, C_i is homologous to C in Y and converges to it in flat norm.

To check the sharp mass bound, let $\vec{T}(x)$ be the oriented unit simple tangent k -vector of T , defined $\|T\|$ -almost everywhere. Define the positive oriented tangent measure V_T by

$$\int \varphi dV_T = \int \varphi(x, \vec{T}(x)) d\|T\|(x),$$

on the product of the compact ambient neighborhood with the space of unit simple k -vectors. Define V_{T_i} in the same way. We claim that

$$V_{T_i} \rightarrow V_T. \quad (16)$$

Indeed, take a weakly convergent subsequence with limit V . Weak convergence of the currents identifies the vector-valued barycenter of V with $\vec{T} \|T\|$. Thus its spatial marginal dominates $\|T\|$. Equality of total masses in (15) makes that marginal exactly $\|T\|$. If ν_x is the conditional probability measure on unit simple vectors, then

$$\int \xi d\nu_x(\xi) = \vec{T}(x), \quad \int \|\xi - \vec{T}(x)\|^2 d\nu_x(\xi) = 0.$$

The second equality uses the Euclidean norm on exterior vectors and $\|\xi\| = \|\vec{T}(x)\| = 1$. Hence $\nu_x = \delta_{\vec{T}(x)}$ almost everywhere. Every subsequential limit is V_T , proving (16).

On each full-dimensional ambient triangulation piece, take the k -Jacobian of the smooth extension of f . For x in the compact neighborhood and a unit simple vector τ , let $J(x, \tau)$ be the maximum of these Jacobians over the finitely many pieces whose closures contain x . The function J is bounded and upper semicontinuous. At $\|T\|$ -almost every x , the tangent vector belongs to the relative open face of K_0 containing x . Every incident piece restricts to $r = \text{id}$ on that face, so its derivative on this tangent agrees with Dh . Consequently,

$$J(x, \vec{T}(x)) = \text{Jac}_k(h|_{\text{Tan}(T, x)}) \quad \text{for } \|T\|\text{-almost every } x.$$

The area formula, (16), and upper semicontinuity give

$$\begin{aligned} \limsup_i \mathbf{M}(C_i) &\leq \limsup_i \int J(x, \tau) dV_{T_i}(x, \tau) \\ &\leq \int J(x, \tau) dV_T(x, \tau) = \mathbf{M}(C). \end{aligned}$$

Flat lower semicontinuity gives the reverse inequality for the limit. This proves the first assertion.

For the filling assertion, refine the triangulation so that the given polyhedral boundary is a simplicial chain. We verify the hypotheses of the homology comparison rather than imposing a mass bound on its filling. A finite Euclidean polyhedron admits locally strong Lipschitz contractions: for a nonempty subset E in a sufficiently small ball, choose $a \in E$ and use

$$(t, x) \mapsto r((1-t)x + ta), \quad x \in E.$$

The intervening line segments lie in U . If L is a local Lipschitz constant for r , the spatial Lipschitz constant is at most L and the time Lipschitz constant is at most $L \text{diam}(E)$. These bounds transfer to Y under the bi-Lipschitz map h . The finite polyhedron is complete, so [8, Corollaries 1.4 and 1.6] apply: the canonical maps identify integral-current homology, Lipschitz singular homology, and singular homology, with integer coefficients. Together with the simplicial-singular homology isomorphism, this identifies the map from simplicial chains to their integration currents as an isomorphism on homology.

A simplicial cycle which is an integral-current boundary therefore has zero simplicial homology class. It is the boundary of a finite integral simplicial chain in the refined triangulation. Mapping that chain by h gives the asserted spherical polyhedral filling, with precisely the prescribed boundary. $\square$

Proof of Lemma 3.1. For part (i), apply Lemma 5.1 to $K = \text{spt } C$ and $Z = \emptyset$. The homotopy gives an integral homology between C and $F_{\#}C$, and (14) controls its mass. If $C = 0$, take $P = 0$; otherwise choose η so that

$$(1 + \eta)^k \mathbf{M}(C) \leq \mathbf{M}(C) + \varepsilon/2.$$

The current $F_{\#}C$ lies in a finite carrier Y and has the same mass in its piecewise metric. Lemma 5.2 gives a homologous spherical polyhedral cycle P in Y with

$$\mathbf{M}(P) \leq \mathbf{M}(F_{\#}C) + \varepsilon/2.$$

The carrier simplices have injective finite-support lifts and the required local isometry property. Thus $P \in \mathcal{P}_k(X_G)$. The two compact integral fillings combine to prove part (i).

For part (ii), let $D \in \mathbf{I}_{k+1}^c(X_G)$ satisfy $\partial D = Z$. Apply Lemma 5.1 to $K = \text{spt } D$, fixing $\text{spt } Z$. Then $F_{\#}D$ is a filling of Z in a finite carrier. Enlarge the carrier if necessary to contain the coordinate spheres of the prescribed boundary, and refine to make its polyhedral pieces a subcomplex. The filling assertion of Lemma 5.2 produces $Q \in \mathcal{P}_{k+1}(X_G)$ with $\partial Q = Z$. $\square$

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